Bernard Benson, PhD

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Research Interests

Machine learning and deep learning for the physical sciences; heliophysics and space-weather forecasting; time-series modeling and uncertainty quantification; natural language processing and large language models (retrieval-augmented generation, agentic and MCP-based tools); computer vision.

Education

Ph.D., Electrical and Computer Engineering
University of Alabama in Huntsville, Huntsville, AL

M.S., Electrical and Computer Engineering
University of Alabama in Huntsville, Huntsville, AL

B.Tech., Electrical and Electronics Engineering
Geethanjali College of Engineering and Technology

Appointments & Professional Experience

Research Scientist VI **May 2025 – Present**
NASA ODSI, University of Alabama in Huntsville, Huntsville, AL

- Developed ensemble large language models for multi-label text classification of scientific documents.
- Built a search wrapper modernizing the backend of NASA's Science Discovery Engine (SDE) using OpenSearch.
- Researched and developed MCP tools for PDS4 and SDE data-search agents.

Generative AI Solutions Architect **Dec 2024 – May 2025**
Megan Soft VA, Remote

- Developed an evaluation framework for generative-AI chatbots and RAG systems using DeepEval and Ragas.
- Established KPIs and testing procedures to optimize performance and user satisfaction.
- Designed multi-chatbot orchestration strategies for agents with overlapping capabilities.
- Designed and implemented recommender systems for pharmacy benefits management.

Data Scientist **Aug 2021 – Nov 2024**
McLeod Software, Birmingham, AL

- Led development and deployment of ML models predicting freight rates for over 1,100 trucking organizations, improving pricing accuracy and efficiency.
- Designed and deployed retrieval-augmented-generation (RAG) chatbots for customer support, reducing resolution times.
- Built and deployed LayoutLMv2-based models to automate document classification.
- Led research and analysis of AI applications across the transportation and logistics industry.
- Represented McLeod Software as a conference speaker at industry events on AI in transportation and logistics.

Applied ML Researcher **Jul 2021 – Aug 2021**
Frontier Development Lab (NASA / SETI Institute), Remote

- Applied machine learning to model the effect of solar drag on satellite orbits as part of a NASA-sponsored research project.

- Developed a multivariate time-series model based on deep neural network ensembles to forecast space-weather indices.
- Earned a team excellence award for the project.

Graduate Teaching Assistant **Aug 2016 – May 2021**

University of Alabama in Huntsville, Huntsville, AL

- Taught and supervised lab sections for CPE211L Introduction to Programming (C++).
- Taught EE316 Electrical Circuits and Electrical Design Lab (Summer 2017).

RF Engineer **Mar 2013 – Mar 2016**

Nexgen Wireless / Global Telecom Associates, Milwaukee, WI

- Analyzed drive-test data and network logs to identify and resolve coverage gaps and interference issues, improving network performance.
- Visualized network performance with heatmaps and coverage plots to support data-driven optimization.

Publications

Refereed Journal Articles

[1] **B. Benson**, W. D. Pan, A. Prasad, G. A. Gary, and Q. Hu, "Forecasting Solar Cycle 25 Using Deep Neural Networks," *Solar Physics*, vol. 295, no. 5, art. 65, 2020.

[2] T. Singh, **B. Benson**, S. A. Z. Raza, T. K. Kim, N. V. Pogorelov, W. P. Smith, and C. N. Arge, "Improving the Arrival-Time Estimates of Coronal Mass Ejections by Using Magnetohydrodynamic Ensemble Modeling, Heliospheric Imager Data, and Machine Learning," *The Astrophysical Journal*, vol. 948, no. 2, art. 78, 2023.

[3] **B. Benson**, W. D. Pan, G. A. Gary, Q. Hu, and T. Staudinger, "Determining the Parameter for the Linear Force-Free Magnetic Field Model with Multi-Dipolar Configurations Using Deep Neural Networks," *Astronomy and Computing*, vol. 26, pp. 50–60, 2019.

[4] T. S. Newman, C. W. Hall, L. Farris, T. Singh, N. Pogorelov, **B. Benson**, S. Raza, *et al.*, "Solar Flare Forecasting Using Machine Learning and SDO/HMI Data: A Multiple Machine Learning Model and Data Curation Technique Comparison Study," *The Astrophysical Journal Supplement Series*, vol. 280, no. 2, art. 54, 2025.

[5] **B. Benson**, W. D. Pan, A. Prasad, G. A. Gary, and Q. Hu, "On the Estimation of the SHARP Parameter MEANALP from AIA Images Using Deep Neural Networks," *Solar Physics*, vol. 296, no. 11, art. 163, 2021.

Conference and Workshop Papers, Abstracts, and Posters

[1] S. Bonasera, G. Acciarini, J. A. Pérez-Hernández, **B. Benson**, E. Brown, *et al.*, "Dropout and Ensemble Networks for Thermospheric Density Uncertainty Estimation," *Bayesian Deep Learning Workshop, NeurIPS 2021*, 2021.

[2] E. J. E. Brown, S. Bonasera, **B. Benson**, J. A. Pérez-Hernández, G. Acciarini, *et al.*, "Learning the Solar Latent Space: Sigma-Variational Autoencoders for Multiple Channel Solar Imaging," *Fourth Workshop on Machine Learning and the Physical Sciences, NeurIPS 2021*, 2021.

[3] **B. Benson**, E. Brown, S. Bonasera, G. Acciarini, J. A. Pérez-Hernández, *et al.*, "Simultaneous Multivariate Forecast of Space Weather Indices Using Deep Neural Network Ensembles," *Fourth Workshop on Machine Learning and the Physical Sciences, NeurIPS 2021*, 2021.

[4] **B. Benson**, Z. Jiang, W. D. Pan, G. A. Gary, and Q. Hu, "Determination of Linear Force-Free Magnetic Field Constant Alpha Using Deep Learning," *International Conference on Computational Science and Computational Intelligence (CSCI)*, 2017.

[5] T. Singh, L. Farris, C. Hall, T. Newman, **B. Benson**, S. Raza, and N. Pogorelov, "Solar Flare Forecasting Using Multiple Machine Learning Models and SDO/HMI Data," *SHINE 2024 Workshop*, 2024.

[6] R. Hooda, W. D. Pan, and **B. Benson**, "Transform Decomposition Switching for Efficient Attribute Compression of 3D Point Clouds Using Neural Networks," *International Conference on Computational Science and Computational Intelligence (CSCI)*, 2022.

[7] T. Singh, **B. Benson**, *et al.*, "Improving the Arrival-Time Prediction of Coronal Mass Ejections Using Magnetohydrodynamic Ensemble Modeling, Heliospheric Imager Data, and Machine Learning," *Proceedings of the 2nd Machine Learning in Heliophysics Conference*, 2022.

[8] C. Davis, B. Praveen, E. Foshee, K. Bugbee, D. Sharma, N. Pantha, S. Awale, **B. Benson**, *et al.*, "LLMs, Encoders, and Agentic Workflows for Curation and Discovery in NASA's Science Discovery Engine," *AGU Fall Meeting 2025*, 2025.

[9] T. Singh, C. Hall, T. Newman, **B. Benson**, S. Raza, and N. Pogorelov, "Solar Flare Forecasting Using Machine Learning and SDO/HMI Data: A Comparison of Multiple ML Models and AR Parameter Time Series," *SHINE 2023 Workshop*, 2023.

Invited Talks

- "Deep Neural Networks Applied to Three Problems in Heliophysics: Coronal Magnetic Field Modeling, Free Magnetic Energy, and Forecasting Solar Activity," NASA Marshall Space Flight Center, Heliophysics Division, Sep. 2020.
- "Applications of AI in Logistics and Supply Chain," Auburn University, Harbert College of Business, Oct. 2023.
- "AI in the Supply Chain," Mississippi State University, College of Business, Mar. 2024.

Teaching Experience

- **CPE211L — Introduction to Programming (C++)**, undergraduate lab instructor, University of Alabama in Huntsville.
- **EE316 — Electrical Circuits and Electrical Design Lab**, undergraduate lab instructor, University of Alabama in Huntsville (Summer 2017).

Honors & Awards

- Team Excellence Award, Frontier Development Lab (NASA / SETI Institute), 2021.

Professional Service

Reviewer

- Research proposals, NASA Heliophysics.
- *Journal of Geophysical Research: Machine Learning and Computation.*
- *Journal of Atmospheric and Solar-Terrestrial Physics.*
- *Solar Physics.*
- Journals of the American Astronomical Society.
- IEEE SoutheastCon — AI and ML tracks (Mar. 2021).

Session Chair

- IEEE SoutheastCon — Computing track (Mar. 2021).

Selected Research Projects

- **Improving CME Arrival Time with Ensemble Models and Machine Learning** — regression and neural-network models to improve coronal mass ejection arrival-time predictions.

- **Simultaneous Multivariate Forecasts of Space-Weather Indices using DNN Ensembles** — LSTM ensembles forecasting space-weather indices over a full solar rotation.
- **Solar Image Processing for Flare Detection** — spatial and frequency-domain filtering to enhance coronal loops, latent-space visualization with VAEs, and relating SHARP parameters to flaring events.
- **Time-Series Forecast of Solar Cycle 25 using Deep Neural Networks** — WaveNet and LSTM models predicting the strength of Solar Cycle 25.

Technical Skills

- **Programming Languages:** Python, SQL, C++, MATLAB.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn; CNNs, RNNs, LSTMs, Transformers; retrieval-augmented generation (RAG) and large language models.
- **Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau.
- **Cloud & DevOps:** AWS, Azure, Google Cloud Platform; Docker, Git, CI/CD pipelines, REST APIs.